



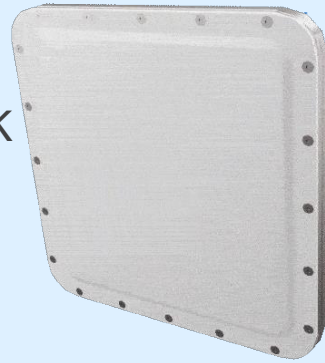
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32x32 RIS

Experimental Validation of Near-Field Beamfocusing using a Reconfigurable Surface

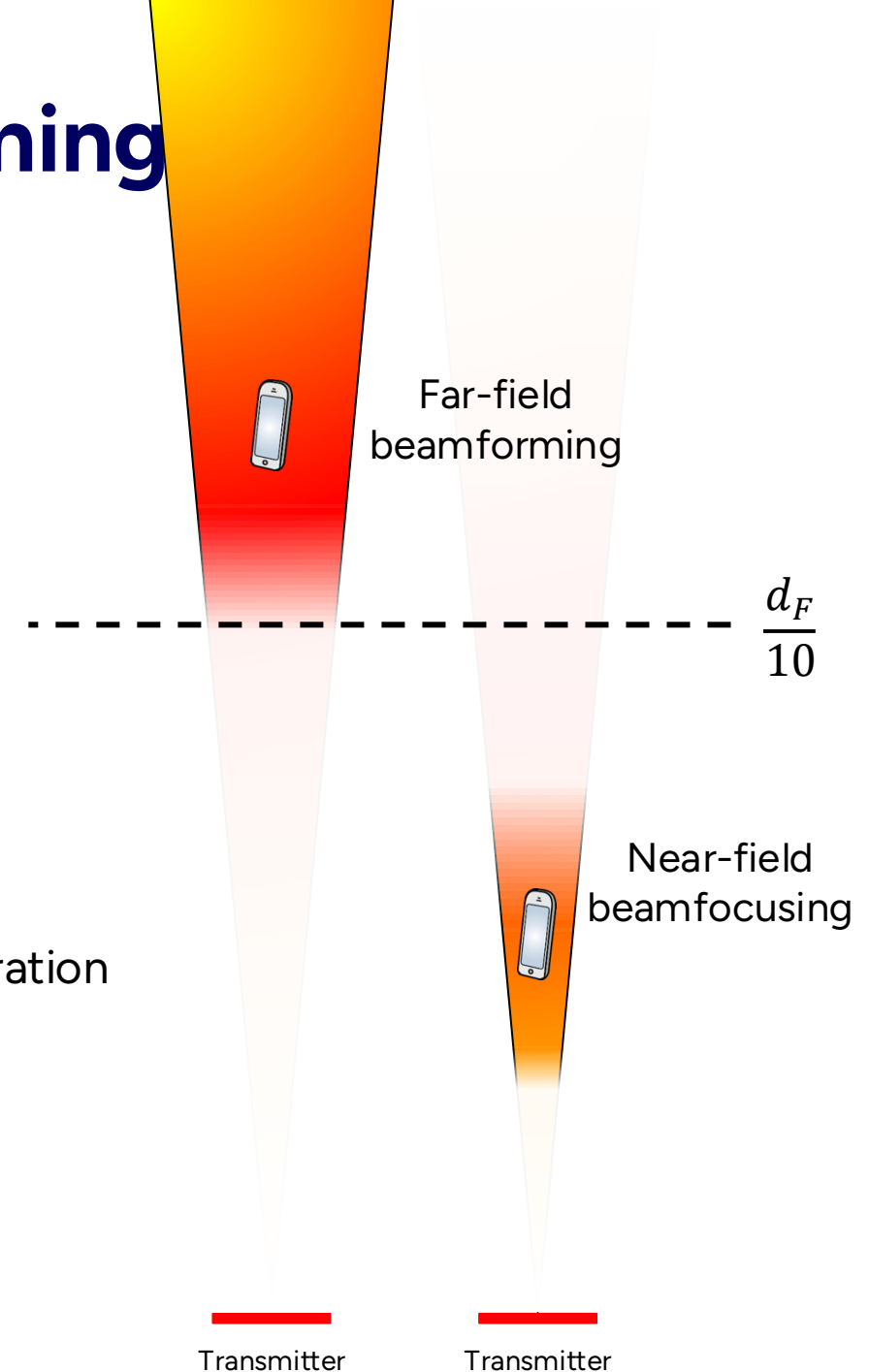
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Beamfocusing vs. Beamforming

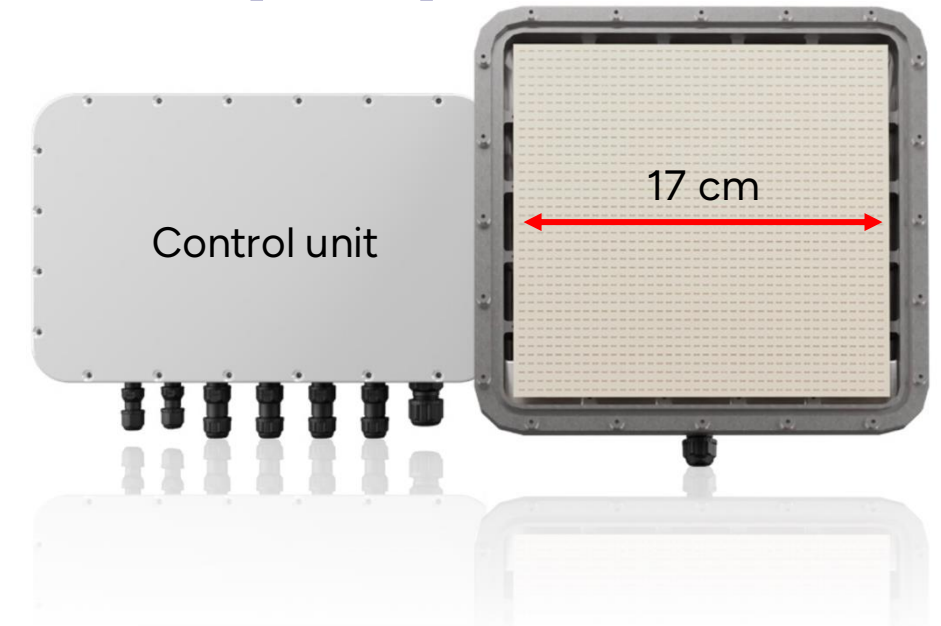
- Radiative near-field zone
 - Grows with aperture length D
 - Shrinks with wavelength λ
- Traditional endpoint: $d_F = \frac{2D^2}{\lambda}$ (Fraunhofer distance)
 - Exact point differs across phenomena
- Finite-depth beamfocusing with square array:
 - Endpoint at $d_F/10$ [12]
 - Assumption: Continuous surface with perfect configuration

Beamfocusing has been experimentally validated for transmitters and fixed metasurfaces



Reconfigurable Intelligent Surface (RIS)

- Experimental setup: RIS at 28 GHz
 - TMYTEK Dynamic RIS
 - 32×32 array with 1-bit phase-shift ($0^\circ, 180^\circ$)
 - Fraunhofer distance: $d_F = \frac{2D^2}{\lambda} = 10.7 \text{ m}$
 - We can observe beamfocusing at realistic distances



Contributions of this paper

First demonstration RIS beamfocusing,
using this array

Generalize beamfocusing theory for b -bit RIS:
Array gain, beamdepth, beamwidth



System Model

- End-to-end channel: $h = \mathbf{h}_r^T \mathbf{\Phi} \mathbf{h}_t$

RIS configuration

$$\mathbf{\Phi} = \text{diag}(e^{j\phi_1}, \dots, e^{j\phi_N})$$

- Near-field array response vector:

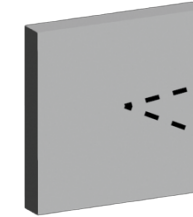
$$\mathbf{a}(\varphi, \theta, d) = \left[e^{-j\frac{2\pi}{\lambda}(d_1-d)}, \dots, e^{-j\frac{2\pi}{\lambda}(d_N-d)} \right]^T$$

- Exact distance for a single array element at $(0, y_n, x_n)$:

$$d_n = \left\| \begin{bmatrix} d \cos(\varphi) \cos(\theta) \\ d \sin(\varphi) \cos(\theta) \\ d \sin(\theta) \end{bmatrix} - \begin{bmatrix} 0 \\ y_n \\ z_n \end{bmatrix} \right\|$$

$$= d \sqrt{1 - \frac{2(\sin(\varphi) \cos(\theta) y_n + \sin(\theta) z_n)}{d} + \frac{y_n^2 + z_n^2}{d^2}}$$

RIS with N elements



$$\mathbf{h}_r = \alpha_r \mathbf{a}(\varphi_r, \theta_r, d_r)$$

Focus point

$$\mathbf{h}_t = \alpha_t \mathbf{a}(\varphi_t, \theta_t, d_t)$$

Complex amplitude

Distance to RIS center

Azimuth angle

Elevation angle

Maximizing end-to-end gain

$$|h|^2 = |\mathbf{h}_r^T \mathbf{\Phi} \mathbf{h}_t|^2 \leq N^2 |\alpha_r \alpha_t|^2$$

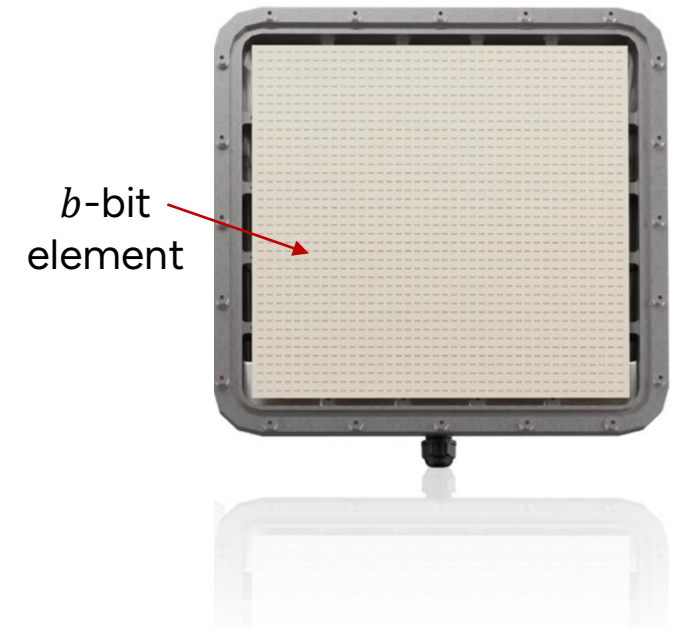
Achieved by $\mathbf{\Phi} = \text{diag}(\mathbf{a}^H(\varphi_t, \theta_t, d_t) \odot \mathbf{a}^H(\varphi_r, \theta_r, d_r))$

Channel Gain with a b -bit RIS

- RIS only supports 2^b phase-shifts per element
 - Quantization to nearest element in $\mathcal{Q}_b = \left[-\frac{\pi}{2^b}, \frac{\pi}{2^b}\right]$
- Quantized RIS configuration:

$$\Phi = \text{diag}(\mathbf{a}^H(\varphi_t, \theta_t, d_t) \odot \mathbf{a}^H(\varphi_r, \theta_r, d_r)) \text{diag}(e^{j\epsilon_1}, \dots, e^{j\epsilon_N})$$

$$\text{Phase errors } \epsilon_n \sim U\left[-\frac{\pi}{2^b}, \frac{\pi}{2^b}\right]$$



Average channel gain:

$$\mathbb{E}\{|h|^2\} = |\alpha_r \alpha_t|^2 (N^2 \text{sinc}^2(2^{-b}) + N (1 - \text{sinc}^2(2^{-b})))$$

$$\text{since } \mathbb{E}\{e^{j\epsilon_n}\} = \text{sinc}(2^{-b})$$

Loss in array gain by a factor
 $\text{sinc}^2(2^{-b}) \in [0.4, 1]$

Averaging motivated by law of large numbers

Same as in [12] as $b \rightarrow \infty$

Quantized Beamfocusing

- What happens to the beam's shape?

Depth of beamfocusing: Focus on distance F instead of d_r

Array gain shrinks from N^2 to

$$\bar{G}(F, d_r) \approx N^2 \text{sinc}^2(2^{-b}) A\left(\frac{d_F \delta}{8}\right) + N(1 - \text{sinc}^2(2^{-b}))$$

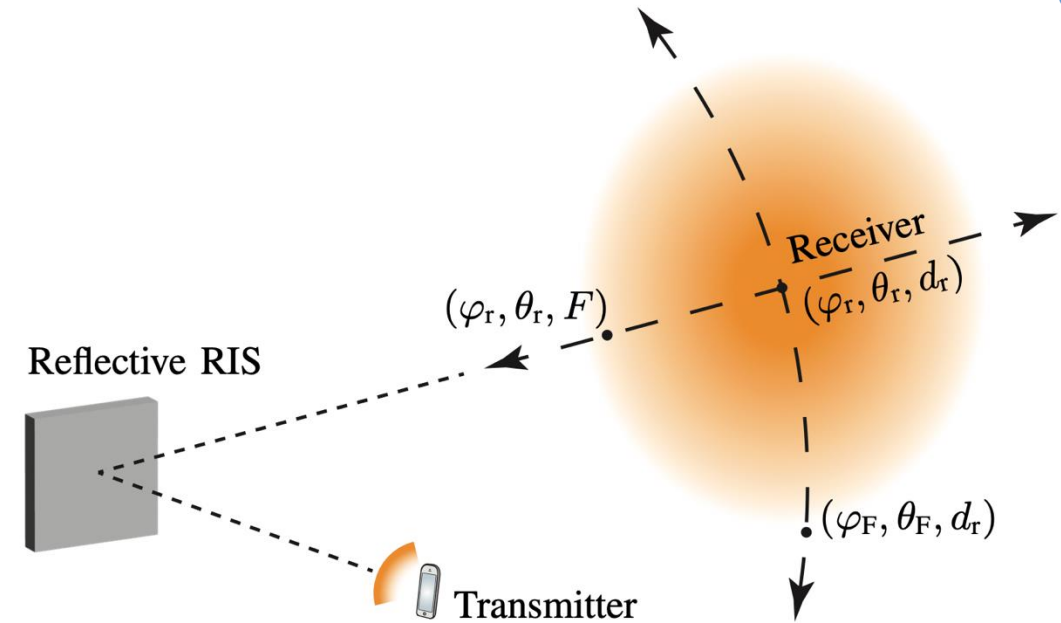
where $\delta = \frac{1}{d_r} - \frac{1}{F}$, $A(x) = \left(C^2(\sqrt{x}) + S^2(\sqrt{x}) \right) / x^2$

Fresnel integrals

Less gain, but almost the same shape gain for any b

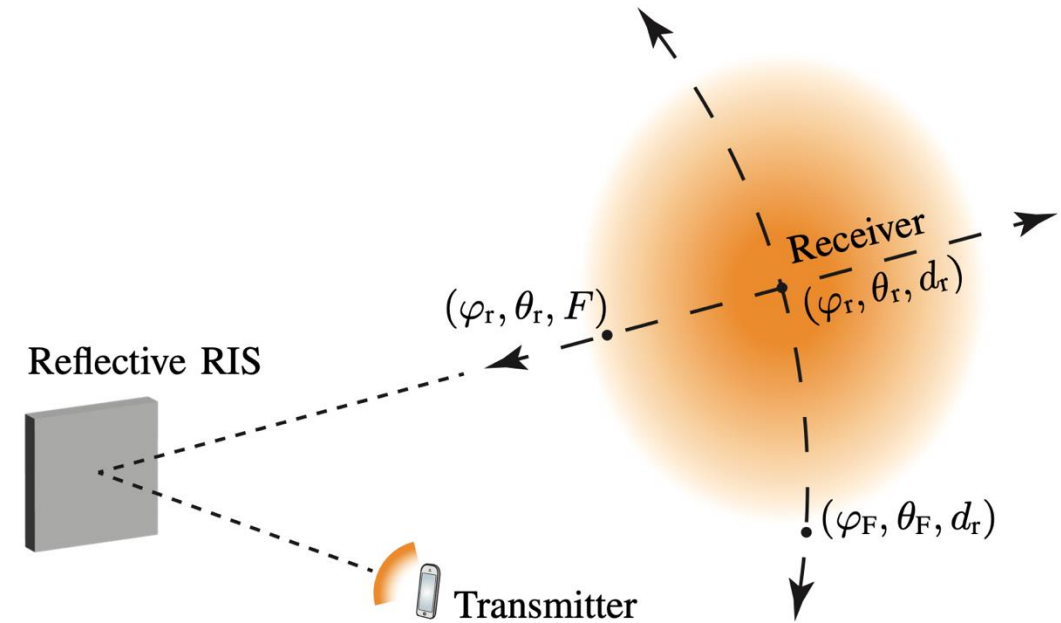
Width of beamfocusing: Focus on wrong angles (φ_F, θ_F)

Analogous result: Less gain but shape beamwidth for any b



Experimental Methodology

- **Purpose:** Compare formulas with measurement
- **Duality:** Moving beam or receiver as same impact
- **Setup:** Everything is fixed, we only move the beam
 - Tests in a real office environment at 28 GHz



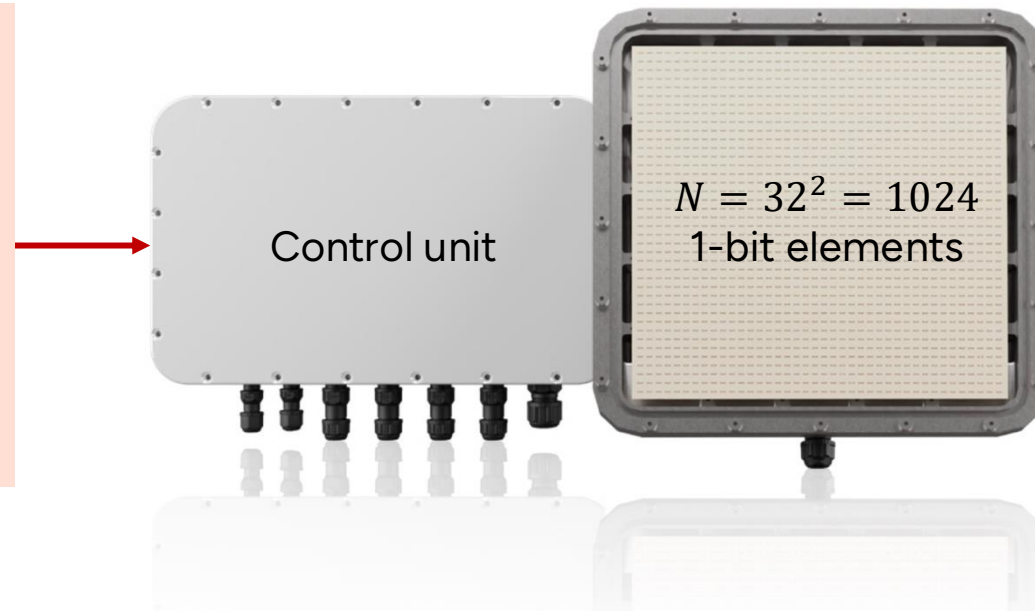
Input:

Transmit location: $(\varphi_t, \theta_t, d_t)$

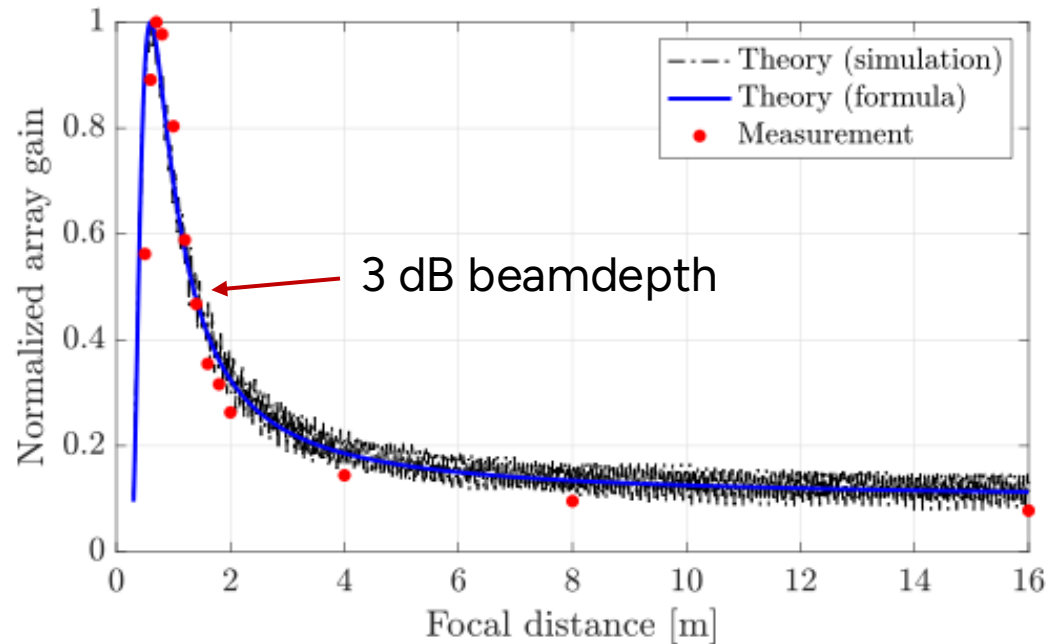
Focus location $(\varphi_r, \theta_r, d_r)$

Software: Sets Φ and compensates for mutual coupling etc.

Output: Received power

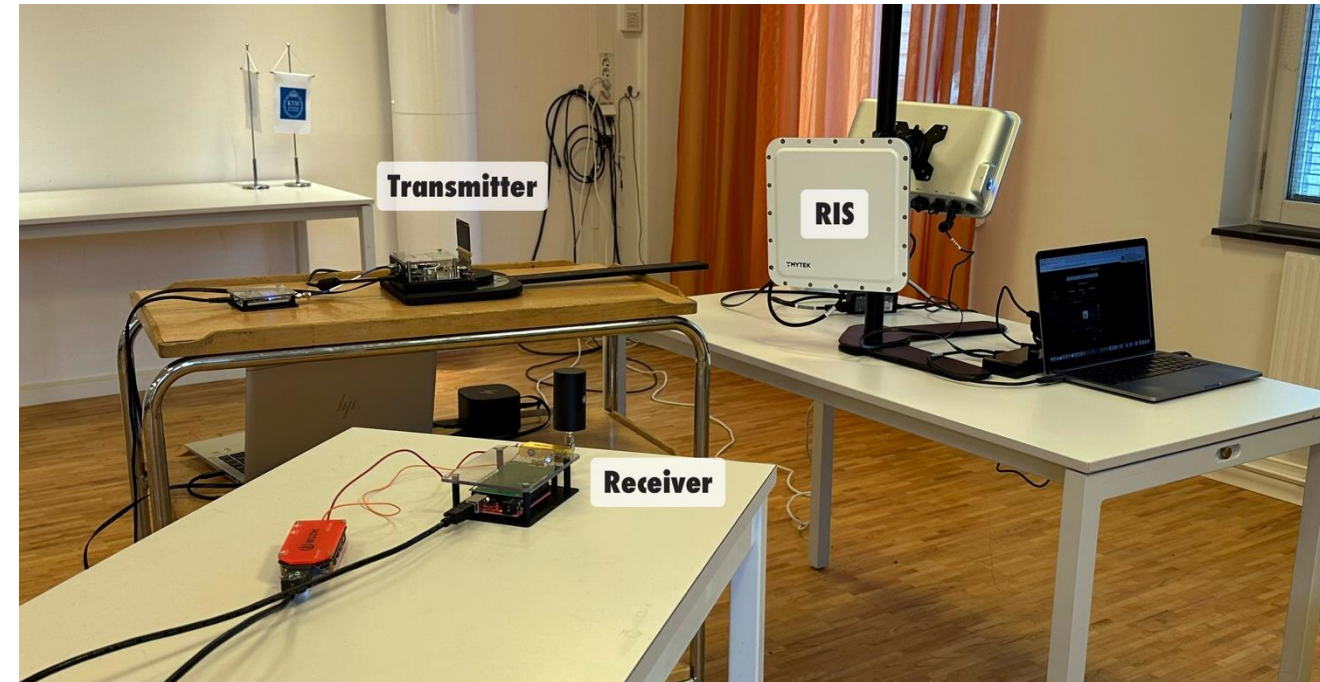


First Experimental Setup: $d_r = 0.6 \text{ m} < d_F/10 = 1.1 \text{ m}$



Theory and measurements match well

Theoretical formulas average out rapid quantization variations

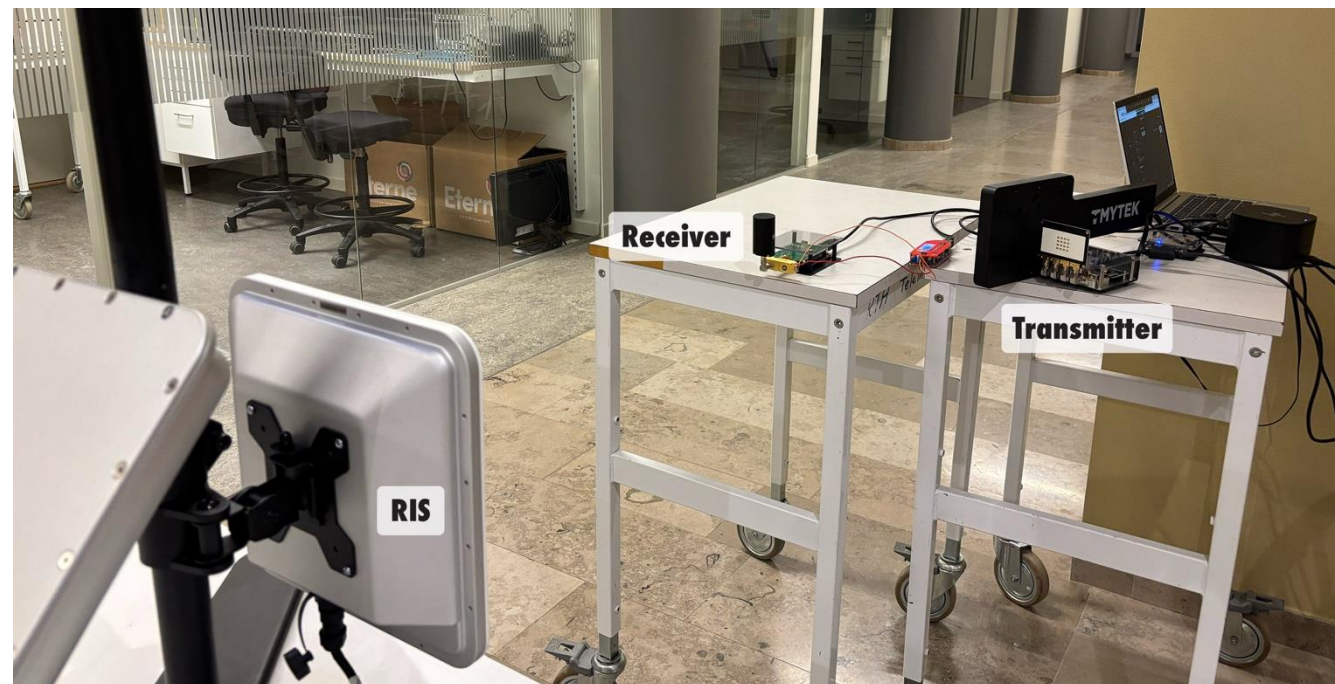
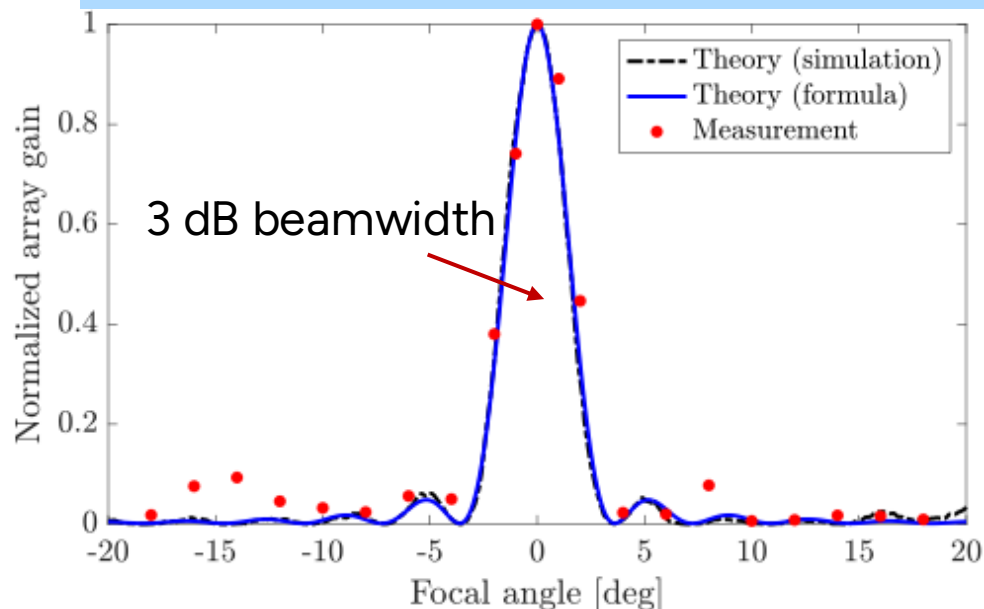
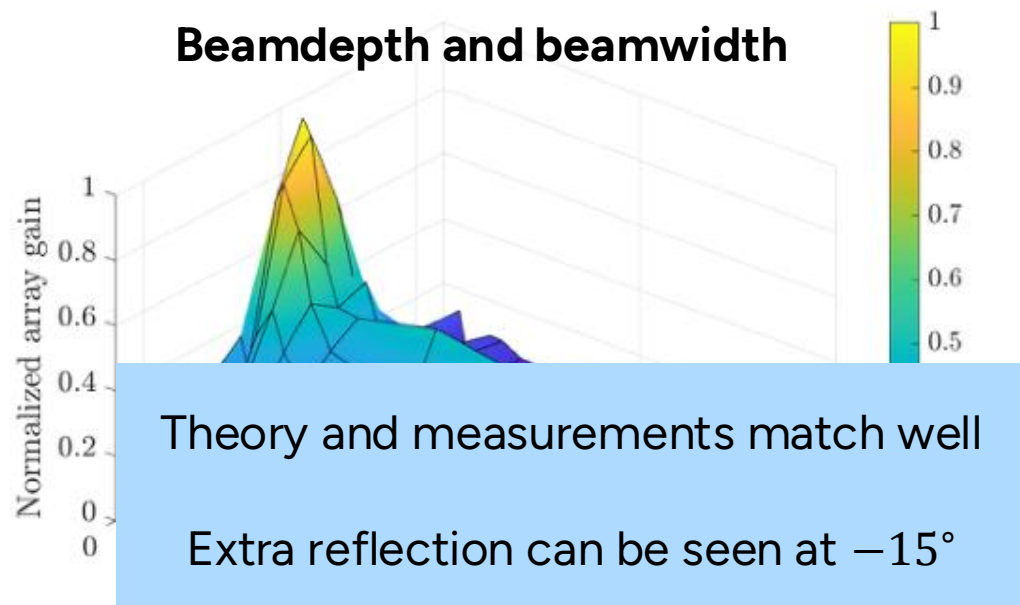


Transmitter: $\varphi_t = -25^\circ, \theta_t = 0^\circ, d_t = 1 \text{ m}$

Receiver: $\varphi_r = 7^\circ, \theta_r = 0^\circ, d_r = 0.6 \text{ m}$

Second Experimental Setup

Beamdepth and beamwidth



Transmitter: $\varphi_t = -17^\circ, \theta_t = 0^\circ, d_t = 1 \text{ m}$

Receiver: $\varphi_r = 0^\circ, \theta_r = 0^\circ, d_r = 0.8 \text{ m}$

Contributions and Conclusions

- **Analytical results:** New beamfocusing formulas for b -bit RIS
- **Experimental results:** First demonstration of beamfocusing with 1-bit RIS
- **Match** between theory and measurements

Conclusions

- Beamfocusing appears at mmWave frequencies in indoor setups
- RIS resolution affects array gain, but not the beam shape
- Formulas that neglect other impairments give accurate results, if the RIS hardware compensates for them

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